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MiMo-VL: From Pre-train to Post-train

Shuhuai Ren
2025/06/14

Xiaomi MiMo

Xiaomi MiMo | LLM Core Team

Chasing AGI
Building powerful, open-sourced, generalist models

2025.4.30

MiMo-7B-Base
MiMo-7B-SFT
MiMo-7B-RL
MiMo-7B-RL-Zero

2025.5.30

MiMo-VL-7B-SFT
MiMo-VL-7B-RL

to be continued...

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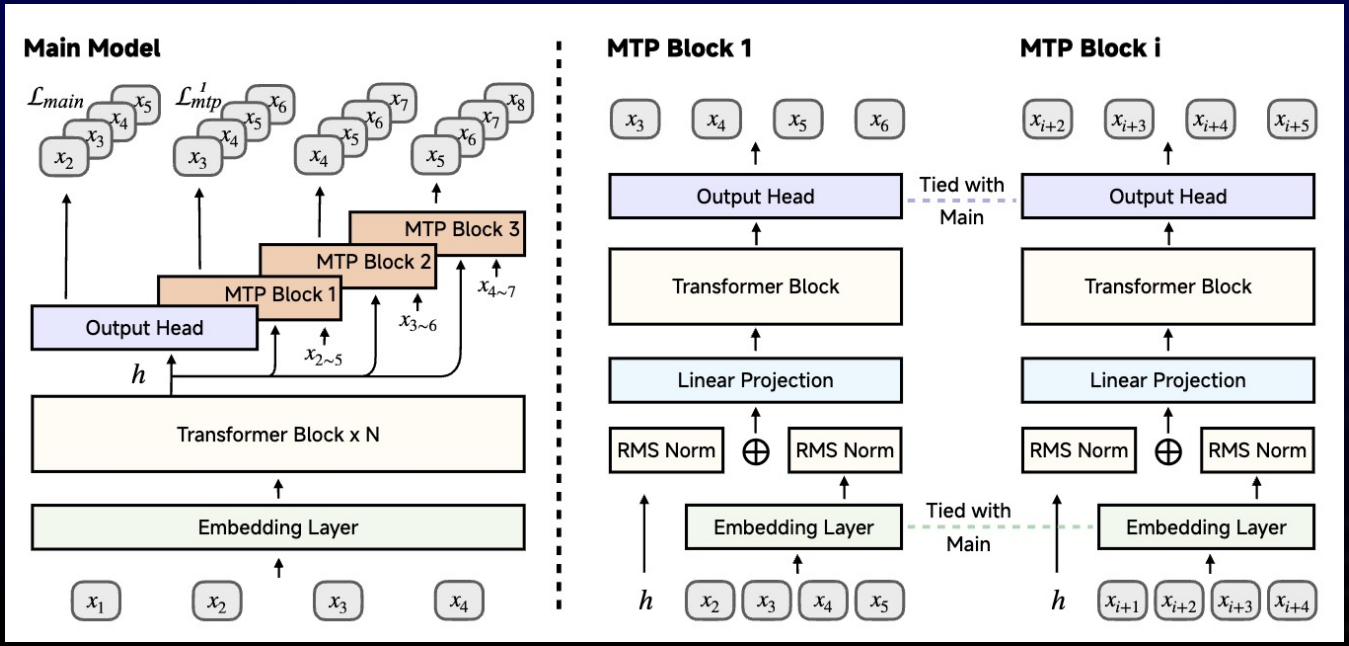
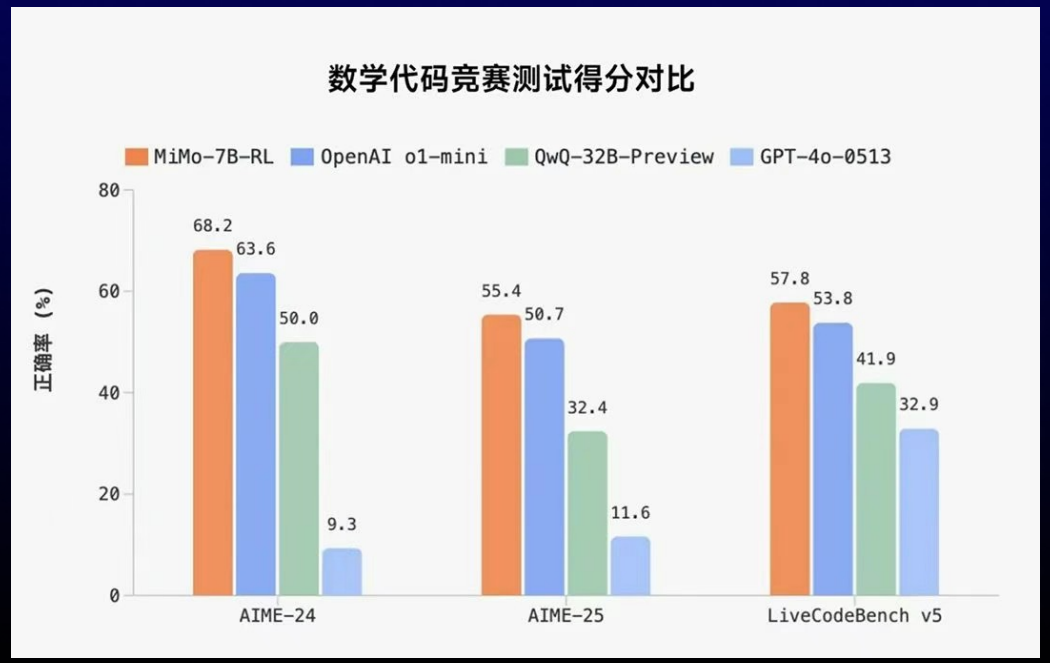
A brief introduction to MiMo-7B



A brief introduction to MiMo-7B



25.4.30

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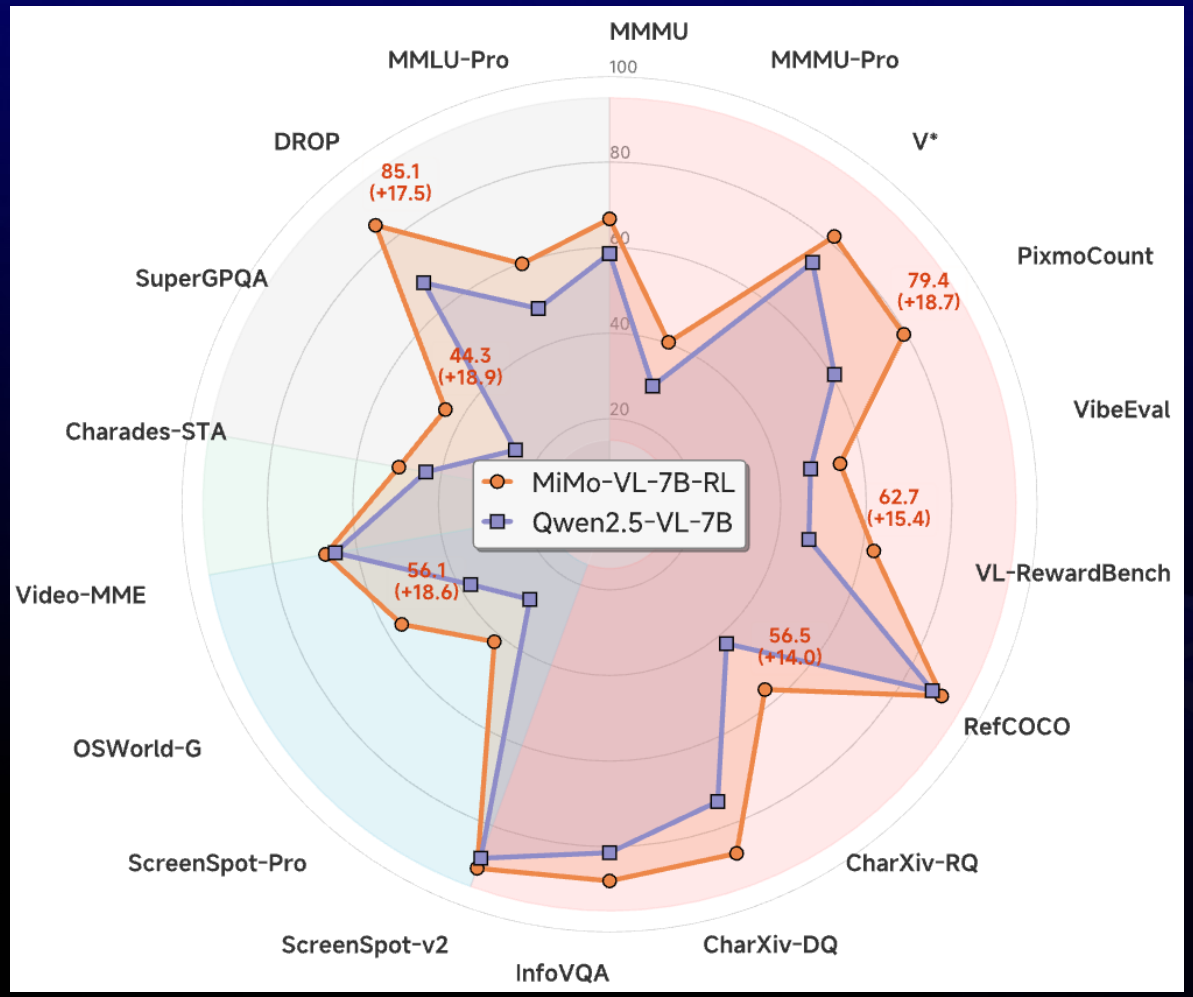
MiMo-VL-7B: Strong Performance



General capabilities: SOTA among open-source models (e.g., Qwen2.5-VL-7B)

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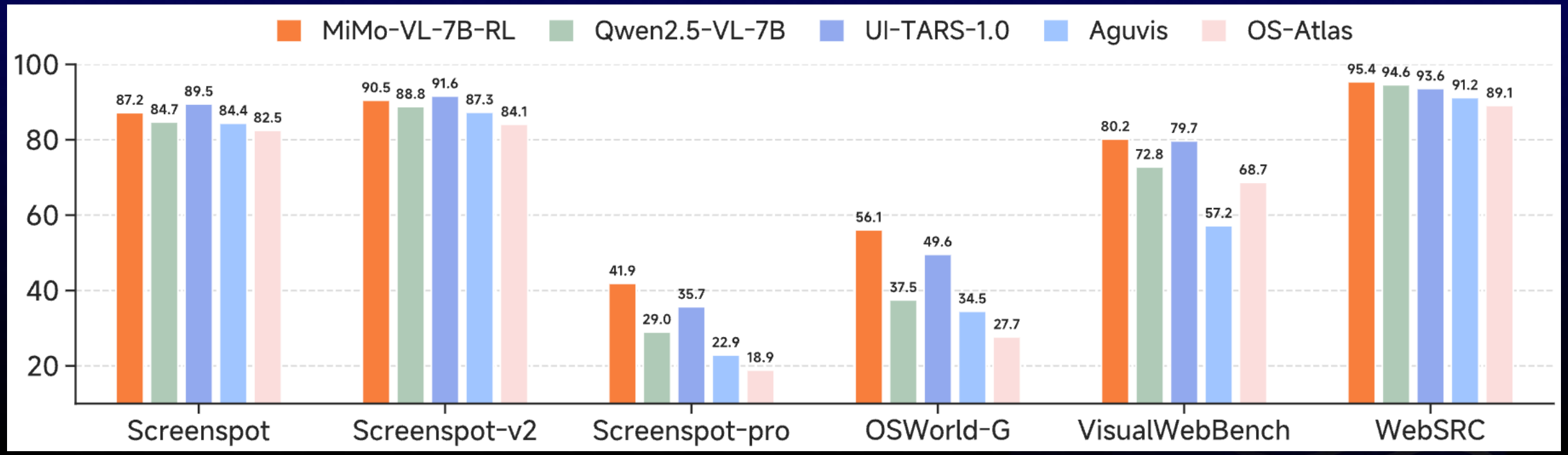
MiMo-VL-7B: Strong Performance



GUI capabilities: on par with specialized models (e.g., UI-TARS)

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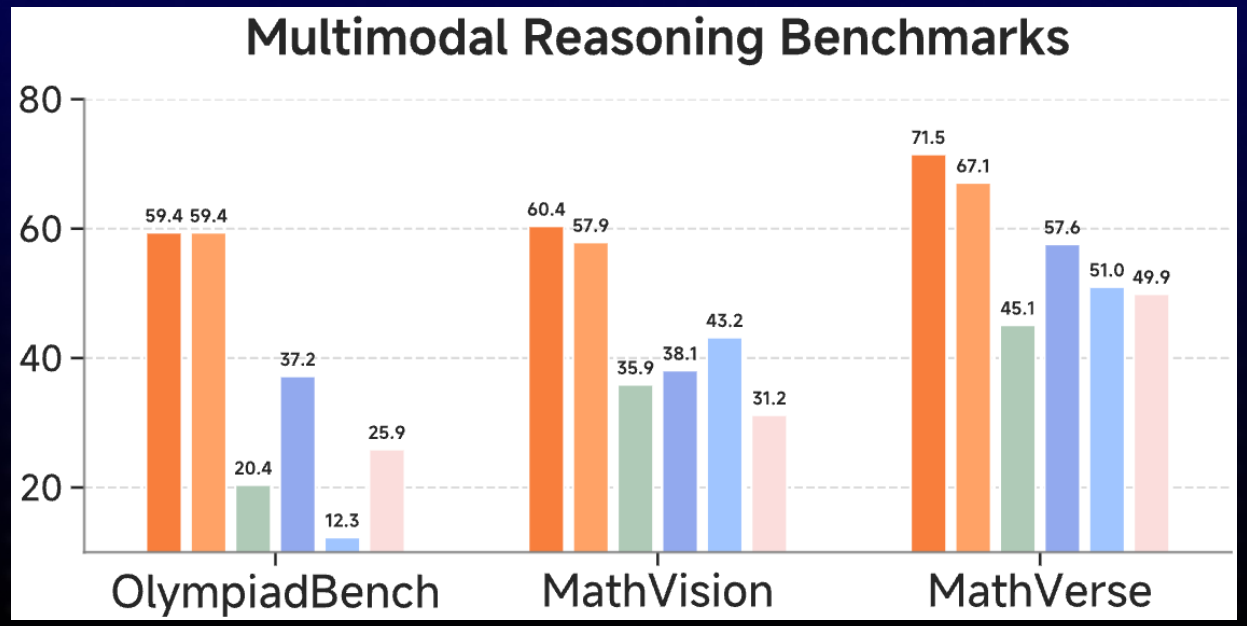
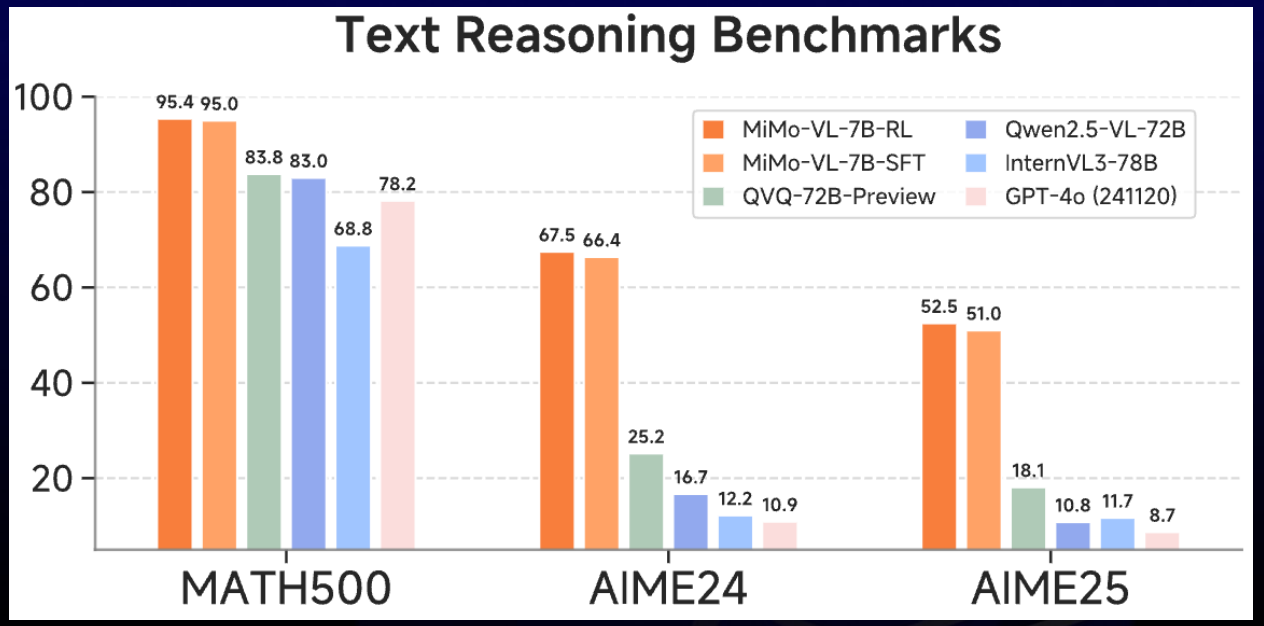
MiMo-VL-7B: Strong Performance





Reasoning capabilities: even outperform Qwen2.5-VL-72B, QVQ-72B-preview, GPT-4o

Cases: scene OCR

Instruction: 这瓶可乐喝10罐会摄入多少克钠?

MiMo-VL-7B-RL: **<think>...</think>**要计算喝10罐可口可乐零度时摄入的钠量，需按以下步骤分析：

1. 步骤**1**：明确营养成分表信息 从图片中“营养成分表”可知：每**100**毫升该饮料中，钠的含量为**18**毫克。

2. 步骤**2**：确定单罐饮料容量（行业常见规格） 可口可乐零度的罐装规格通常为**330**毫升/罐（这是饮料行业的常见罐装容量，若实际容量不同需以产品标注为准，此处默认330ml）。

3. 步骤**3**：计算单罐钠含量 单罐（330ml）中钠的含量= 每100ml钠含量× 单罐容量÷ 100 即：18 毫克/100ml × 330 ml ÷ 100 = 59.4 毫克

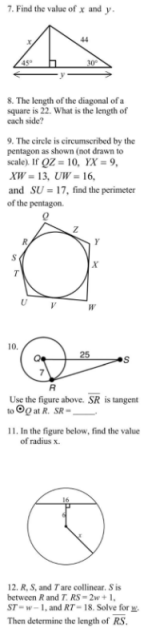
4. 步骤**4**：计算**10**罐钠总量 10罐钠总量= 单罐钠含量× 10 即：59.4 毫克 × 10 = 594 毫克

因此，喝10罐该款可口可乐零度，约摄入**594**毫克 钠。

Cases: detailed perception problem

Instruction: How many muffin in the image?
MiMo-VL-7B-RL: <think>...</think> To determine the number of muffins in the image, we analyze the 3×4 grid (12 cells total) and count muffins row by row:
- **Top row (row 1)**: Columns 1 and 3 are muffins → 2 muffins. - **Middle row (row 2)**: Columns 2 and 4 are muffins → 2 muffins. - **Bottom row (row 3)**: Columns 1 and 3 are muffins → 2 muffins.
Adding these, $2 + 2 + 2 = 6$.
There are 6 muffins in the image.

Cases: multiple math problems



Instruction: answer all the questions.

MiMo-VL-7B-RL: <think>...</think>

1. Problem 7 To find x and y, we analyze the two right triangles formed by the height:

   - **Right triangle (30° angle)**: Hypotenuse $= 44$. The height h (opposite 30°) is $44 \cdot \sin(30°) = 44 \cdot 0.5 = 22$. The base segment y (adjacent to 30°) is $44 \cdot \cos(30°) = 44 \cdot \frac{\sqrt{3}}{2} = 22\sqrt{3}$.

   - **Right triangle (45° angle)**: This is a 45-45-90 triangle, so legs are equal. The height $h = 22$, so the other leg (left base segment) is also 22. Hypotenuse x (45-45-90 ratio: leg $\cdot\sqrt{2}$) is $22\sqrt{2}$.

2. Problem 8 For a square with diagonal $d = 22$, the side length s relates to the diagonal by $d = s\sqrt{2}$. Solving for s: $s = \frac{d}{\sqrt{2}} = \frac{22}{\sqrt{2}} = 11\sqrt{2}$ (rationalizing the denominator).

3. Problem 9 A pentagon circumscribed about a circle (tangential pentagon) has the property that the sum of the lengths of all sides equals 2× the sum of the tangent segments from each vertex. However, since the problem provides the side lengths $QZ = 10$, $YX = 9$, $XW = 13$, $UW = 16$, $SU = 17$, and the pentagon is tangential, the perimeter is the sum of these sides (as they represent the sides of the pentagon).

   Perimeter $= 10 + 9 + 13 + 16 + 17 = 65$.

4. Problem 10 $\overline{SR}$ is tangent to $\odot Q$ at R, so $QR \perp SR$ (tangent is perpendicular to radius). Thus, $\triangle QRS$ is a right triangle with $QR = 7$ (radius) and $QS = 25$ (hypotenuse). By the Pythagorean theorem: $SR = \sqrt{QS^2 - QR^2} = \sqrt{25^2 - 7^2} = \sqrt{625 - 49} = \sqrt{576} = 24$

5. Problem 11 A perpendicular from the circle's center to a chord bisects the chord. The chord length is 16, so half the chord is $\frac{16}{2} = 8$. The distance from the center to the chord is 6, forming a right triangle with legs 6 and 8, and hypotenuse x (radius). By the Pythagorean theorem: $x = \sqrt{6^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = 10$

6. Problem 12 Since R, S, T are collinear and S is between R and T, $RS + ST = RT$. Substitute $RS = 2w + 1$, $ST = w - 1$, and $RT = 18$:

   $$(2w + 1) + (w - 1) = 18$$

   Simplify: $3w = 18 \implies w = 6$. Substitute $w = 6$ into RS: $RS = 2(6) + 1 = 13$

Final Answers 7. $x = 22\sqrt{2}$, $y = 22\sqrt{3}$ 8. $11\sqrt{2}$ 9. 65 10. 24 11. 10 12. $w = 6$, $\overline{RS} = 13$

Input image:

Instruction: 完成该试卷

MiMo-VL-7B: <think>...</think>

第1题：集合的交集 集合$A = \{0, 1, 2, 8\}$，$B = \{-1, 1, 6, 8\}$，交集为同时属于两个集合的元素，即$A \cap B = \{1, 8\}$。

第2题：复数的实部 设$z = x + yi$（x, y 为实数），由$i \cdot z = 1 + 2i$，即$i(x + yi) = -y + xi = 1 + 2i$。分离实部和虚部：$-y = 1$（实部），$x = 2$（虚部）。故$z$ 的实部为2。

第3题：茎叶图的平均数 茎叶图中，茎为十位，叶为个位。分数为$89, 89, 90, 91, 91$（共5个）。平均数为$\frac{89+89+90+91+91}{5} = \frac{450}{5} = 90$。

第4题：伪代码的循环结果 初始化$I = 1$，$S = 1$。循环条件$I < 6$：- 第1次循环：$I = 1 + 2 = 3$，$S = 2 \times 1 = 2$；- 第2次循环：$I = 3 + 2 = 5$，$S = 2 \times 2 = 4$；- 第3次循环：$I = 5 + 2 = 7$（不满足$I < 6$，退出循环）。最终$S = 8$。

第5题：函数的定义域 函数$f(x) = \sqrt{\log_2 x - 1}$ 需满足：1. 二次根式内非负：$\log_2 x - 1 \geq 0$；2. 对数有定义：$x > 0$。由$\log_2 x \geq 1$ 得$x \geq 2^1 = 2$，故定义域为$[2, +\infty)$。

第6题：组合的概率 2男3女中选2人，总选法$C_5^2 = 10$；选中2女生的选法$C_3^2 = 3$。概率为$\frac{3}{10}$。

第7题：正弦函数的对称轴 函数$y = \sin(2x + \varphi)$ 关于$x = \frac{\pi}{3}$ 对称，即$2 \cdot \frac{\pi}{3} + \varphi = k\pi + \frac{\pi}{2}$（$k \in \mathbb{Z}$）。取$k = 0$，得$\varphi = -\frac{\pi}{6}$（满足$-\frac{\pi}{2} < \varphi < \frac{\pi}{2}$）。

第8题：双曲线的离心率 双曲线$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$，右焦点$F(c, 0)$（$c = \sqrt{a^2 + b^2}$），渐近线$bx - ay = 0$。焦点到渐近线的距离为$\frac{|bc|}{\sqrt{a^2+b^2}} = b$，由题设$b = \frac{\sqrt{3}}{2}c$。结合$c^2 = a^2 + b^2$，得$a = \frac{c}{2}$，离心率$e = \frac{c}{a} = 2$。

第9题：周期函数与复合函数 $f(x + 4) = f(x)$ 说明周期为4。- $f(15) = f(15 - 4 \times 3) = f(3) = f(-1)$（$3 = -1 + 4$）；- $x \in (-2, 0]$ 时，$f(-1) = |-1 + \frac{1}{2}| = \frac{1}{2}$；- $f(\frac{1}{2}) = \cos\left(\frac{\pi}{2} \cdot \frac{1}{2}\right) = \cos\frac{\pi}{4} = \frac{\sqrt{2}}{2}$（$\frac{1}{2} \in (0, 2]$）。

第10题：多面体的体积 正方体棱长2，面中心坐标为$(\pm1, 0, 0), (0, \pm1, 0), (0, 0, \pm1)$，构成正八面体。正八面体体积为两个四棱锥之和，底面正方形边长$\sqrt{2}$，高1，体积为$2 \times \frac{1}{3} \times 2 \times 1 = \frac{4}{3}$。

第11题：函数的极值与零点 函数$f(x) = 2x^3 - ax^2 + 1$在$(0, +\infty)$内有且只有一个零点，导数$f'(x) = 6x^2 - 2ax$，临界点$x = 0$ 和$x = \frac{a}{3}$。当$f(\frac{a}{3}) = 0$ 时（极小值为0，仅1个零点），解得$a = 3$。$f(x)$ 在$[-1, 1]$ 上，$f(0) = 1$（最大值），$f(-1) = -4$（最小值），和为$1 + (-4) = -3$。

第12题：向量垂直的条件 设$A(x, 2x)$，$AB = (5 - x, -2x)$，圆心$C\left(\frac{x+5}{2}, x\right)$，另一交点$D(1, 2)$（由圆与直线交点唯一性得）。向量$\overrightarrow{CD} = \left(\frac{-x-3}{2}, 2 - x\right)$，由$\overrightarrow{AB} \cdot \overrightarrow{CD} = 0$，解得$x = 3$。

答案

1. $\{1, 8\}$ 2. **2** 3. **90** 4. **8** 5. $[2, +\infty)$ 6. $\frac{3}{10}$ 7. $-\frac{\pi}{6}$ 8. **2** 9. $\frac{\sqrt{2}}{2}$ 10. $\frac{4}{3}$ 11. **−3** 12. **3**

各家大模型

在“2025 年数学新课标 I 卷”得分对比

7B小模型追平235B Qwen3

模型	模型参数量	客观题得分（满分73分）	解答题得分（满分77分）	总分（满分150分）
Gemini 2.5 pro	总参>1T（预估）	68	77	145
Doubao-1.5-thinking-vision-pro	总参532B，激活20B	68	76	144
DeepSeek R1	总参671B，激活37B	68	76	144
OpenAI o3	总参>500B（预估）	65	75	140
MiMo-VL-7B	总参7B	68	71	139
Qwen3-235B	总参235B，激活22B	68	71	139
hunyuan-t1-latest	总参72B	68	68	136
文心X1 Turbo	总参>100B（预估）	68	66	134
Qwen2.5-VL-7B	总参7B	50	33	83

超同参数量Qwen2.5-VL-7B整56分！

Cases: GUI tasks

Task Setup

You are a GUI Agent. I want to purchase a Xiaomi SU7 on the Xiaomi official website. Please plan the actions briefly. When your action triggers a change in the interface, I will provide you with a new screenshot of the interface, and you need to think and then predict the next action in JSON format.

Step 1

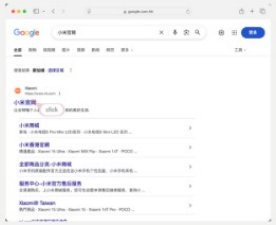
User: First, search the official website of Xiaomi in the search Bar

Model:
- <think> The user requires to search the official website of Xiaomi in google search bar. First look at the screenshot, the search bar is where we input keywords. So the action is to click search bar and enter keywords.</think>
- Action: { "action": "click", "start_point": [657, 639]}

Step 2

Model:
- Action: { "action": "input", "text": "小米官网"}

Step 3



Model:
- Action: { "action": "click", "start_point": [204, 697]}

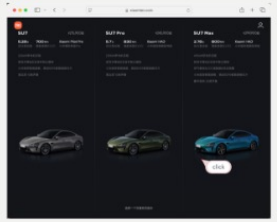
Step 4

Model:
- Action: { "action": "click", "start_point": [937, 1181]}

Step 5

Model:
- Action: { "action": "click", "start_point": [1201, 1347]}

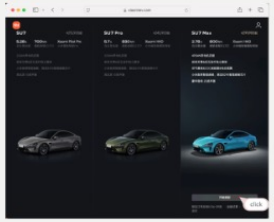
Step 6



User: Ok, it's the current stage. I would like to buy SU7 max.

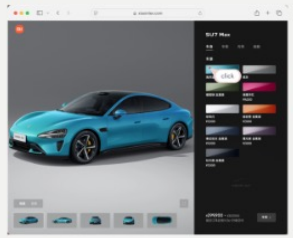
Model:
- <think>The user wants to buy SU7 Max. First find the place to select version, then click the right version. In this way, I can continue to buy. So the action is click SU7 Pro. </think>
- Action: {"action": "click", "start_point": [1394, 1123]}

Step 7



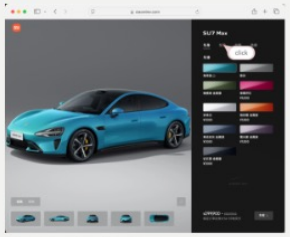
Model:
- Action: { "action": "click", "start_point": [1677, 1389]}

Step 8



Model:
- Action: { "action": "click", "start_point": [1362, 420]}

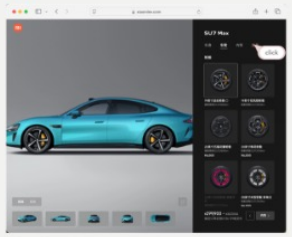
Step 9



User: Now, first describe elements in current screenshot, then give the next action.

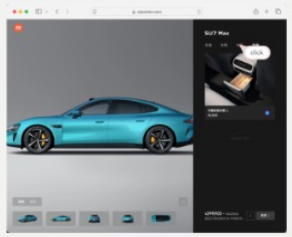
Model:
- <think> [….] </think>
- On the left, there is a frontal view of the Bay Blue SU7 Max vehicle, with thumbnails of the vehicle from different angles at the bottom, and buttons for "View" and "Take Photo" below. On the right side is the configuration options area for the SU7 Max, currently on the "Body" configuration page, displaying various paint color options along with their corresponding prices. […]
- Action: {"action": "click", "start_point": [1475, 276]}

Step 10



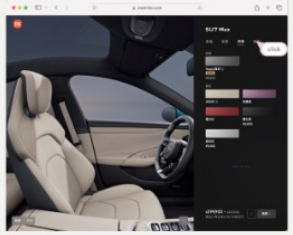
Model:
- <think>The user is now in the wheels configuration interface and needs to continue configuring optional features. The next step should be to click the "Optional Features" tab to enter the optional content page, allowing for further selection of other accessories. Therefore, the click command should be sent to the optional features tab. </think>
- Action: {"action": "click", "start_point": [1672, 276]}

Step 11



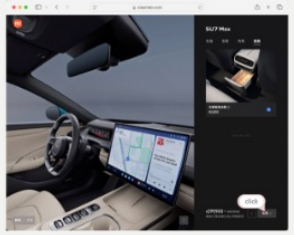
Model:
- Action: {"action": "click", "start_point": [1569, 276]}

Step 12



Model:
- Action: {"action": "click", "start_point": [1675, 276]}

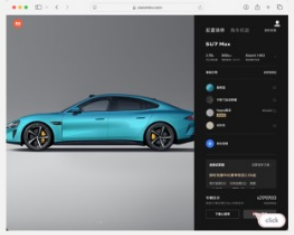
Step 13



User: Click overview, get ready to buy the car.

Model:
- Action: {"action": "click", "start_point": [1728, 1421]}

Step 14



Model:
- Action: {"action": "click", "start_point": [1657, 1423]}

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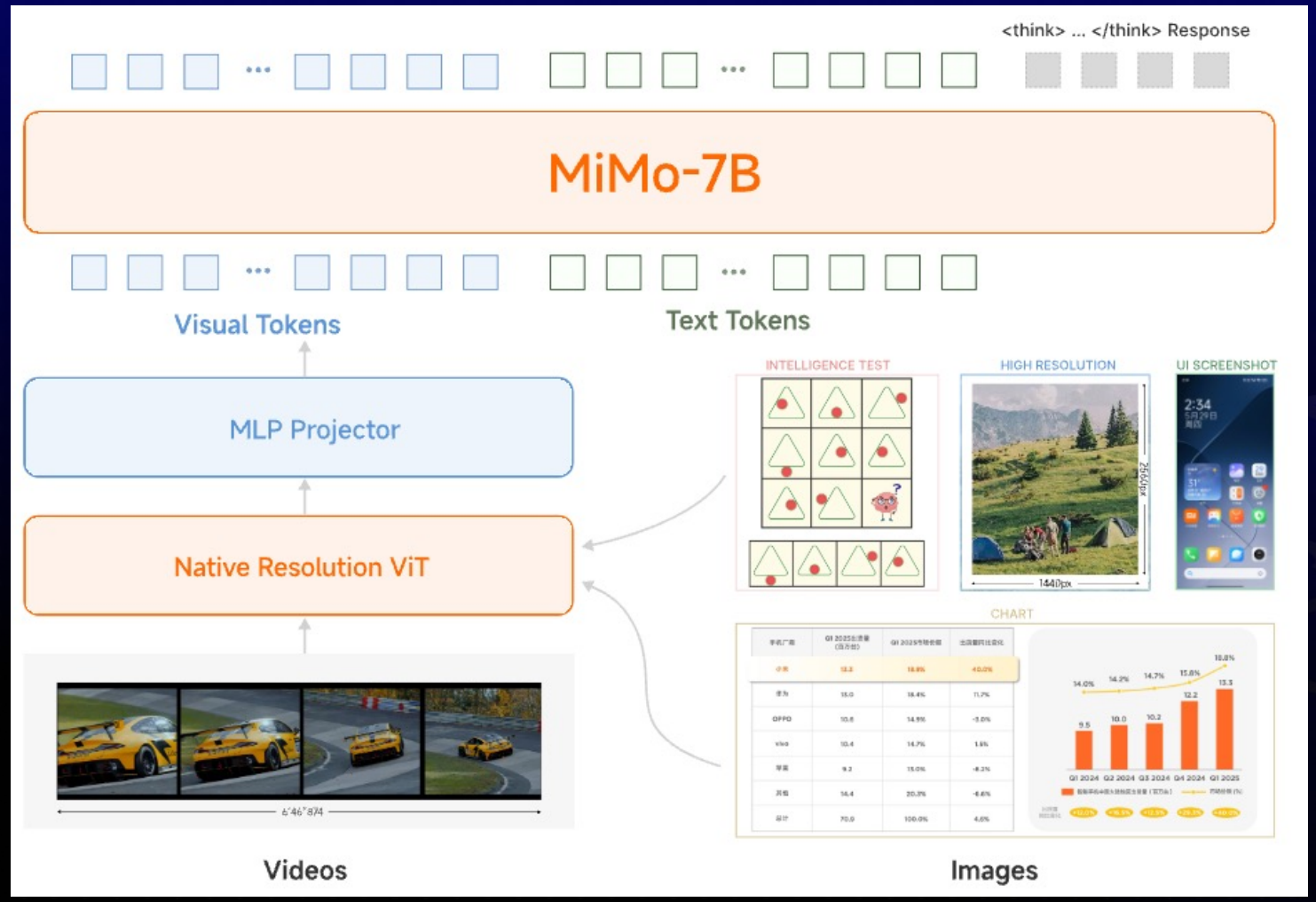
Architecture



Pre-training

Stages	Stage 1	Stage 2	Stage 3	Stage 4
Purpose	Projector Warmup	Vision-Language Alignment	Multimodal Pre-training	Long-context SFT
Dataset	Image-Caption Pairs	+ Interleaved Data	+ Pure Text, OCR, Grounding, QA, Video, GUI, Instruction Data, Reasoning Data	+ Long Pure Text, Long Documents, High-resolution Images, Extended Videos, Long Reasoning Data
Learning Rate	1e-3	1e-4, 1e-5	1e-5	2.5e-5
Training Tokens	300B	167B	1.4T	550B
Sequence Length	8K	8K	8K	32K
Trainable Components	Projector	ViT, Projector	All	All

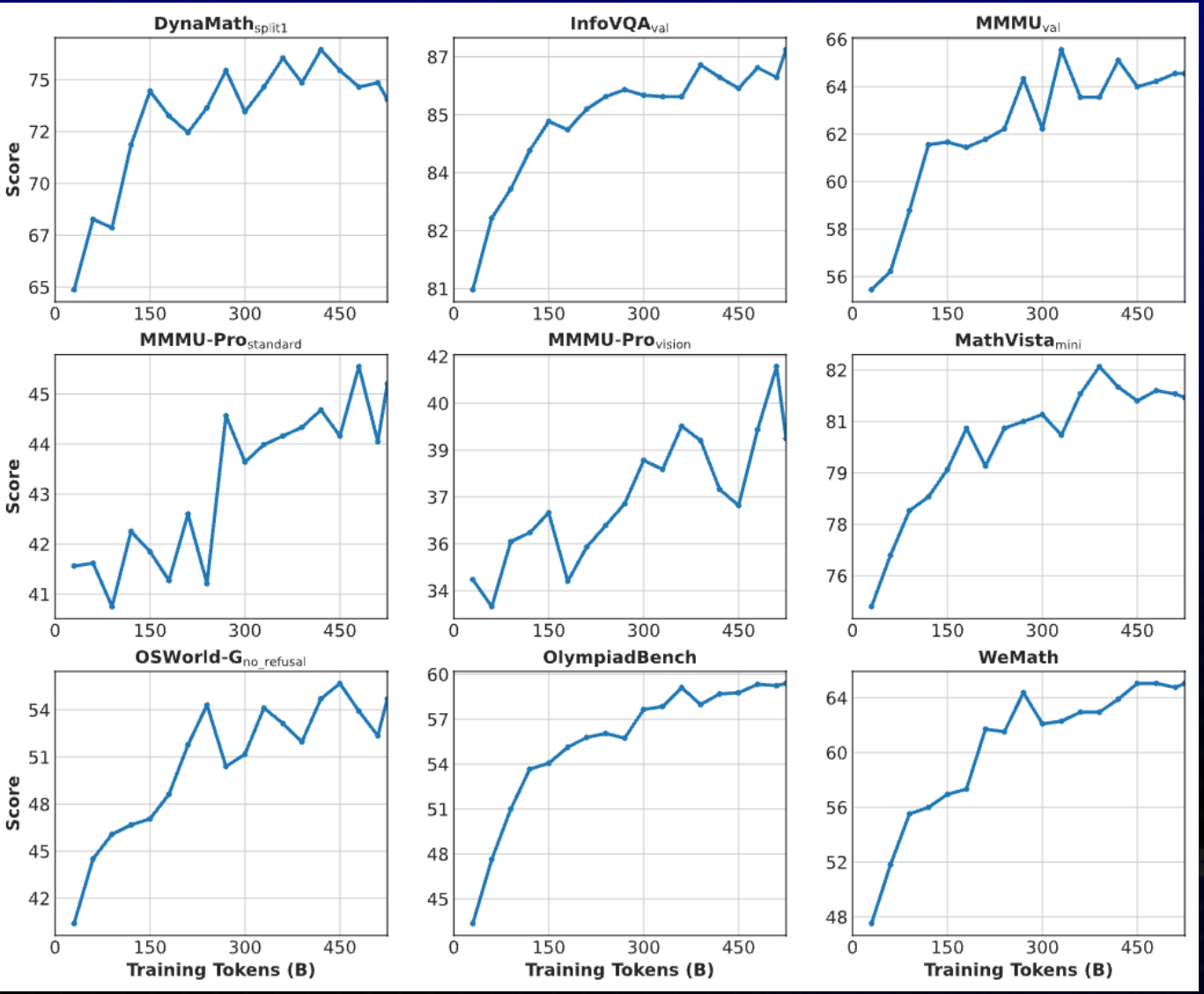
Overview of MiMo-VL-7B training stages

Long reasoning data synthesis

- Collect large volume of diverse, high-quality querise
  - Perceptual QA, document QA, video QA, visual reasoning...
- Synthesize response with thinking process
- Format matters
  - <think>...</think> ... \boxed{}
  - Every special tag is correct?
  - <think> and </think> are paired?
  - Thinking process is repeated all the time?
  - Thought is clear?
- Long-context matters

Stage 4 training

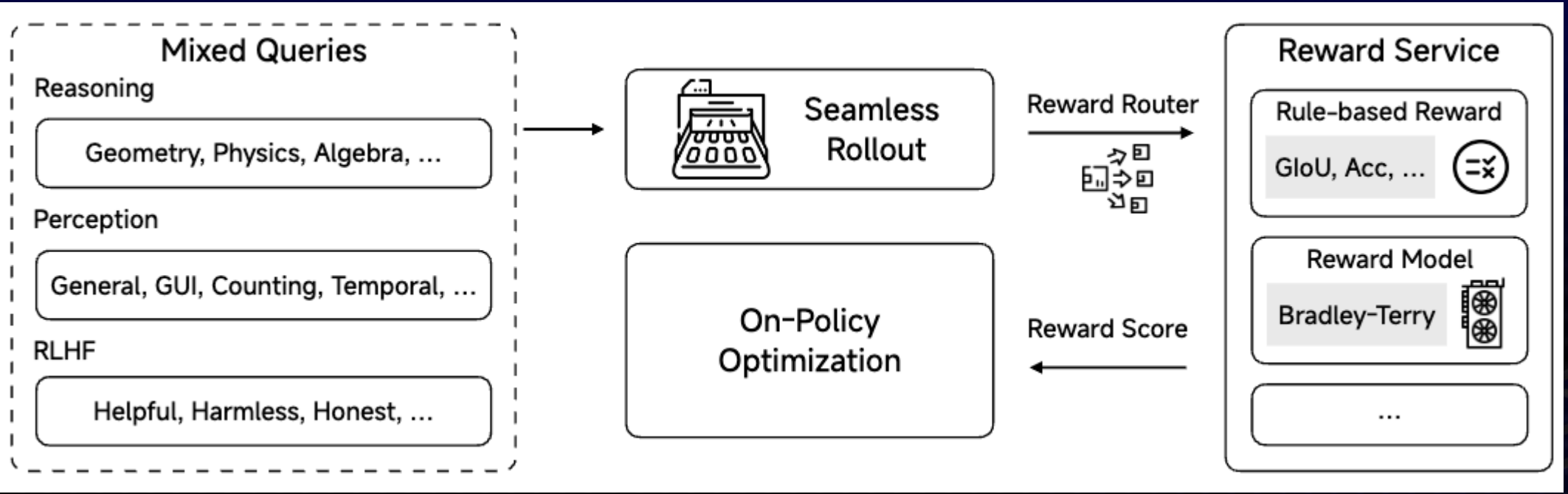
- Distinct from "Lite SFT", we use "Heavy SFT"
- For the large volume of traditional short-answer QA data, we largely discontinued its use
- Performance continued to improve over multiple epochs, demonstrating resilience against saturation



Reasoning capability significantly improved during stage 4

Post-training

MORL: Mixed On-policy RL



MORL: Mixed On-policy RL

- RLVR
  - Reasoning
    - Visual Reasoning
    - Text Reasoning
  - Perception
    - Image (GUI) Grounding
    - Visual Counting
    - Temporal Video Grounding
- RLHF

MORL: Mixed On-policy RL

- RLVR
  - Reasoning
    - Compile diverse verifiable STEM questions
    - Filter proof-based problems
    - Rewrite multiple-choice questions into free-answer formats
    - Exclude questions that are either too hard or too easy
    - Data cleaning and category balancing
    - Reward: rule-based Math-Verify

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- RLVR
  - Perception
    - Reward: GIoU / IoU / Acc
    - Need to define too-hard / too-easy samples, be careful about passrate
    - Do not need format reward if already trained with cold-start data

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- RLHF
  - Gather multimodal and text-only queries from open-source instruction tuning datasets and in-house human-written source
  - Cluster to avoid homogeneous queries
  - Balance the proportions of Chinese and English queries, as well as helpfulness and harmlessness
  - Use multiple top-performing VLMs to generate responses
  - Rank responses with an advanced VLM
  - Train reward model (based on MiMo-7B and MiMo-VL-7B)
  - Use the same query set for both reward model training and the RLHF process

MORL: Mixed On-policy RL

- On-Policy RL Recipe
  - Single-step policy updates following response rollout
  - Removal of the KL loss
  - Dynamic sampling
  - Easy data filter
  - Re-sampling strategies
- Reward-as-a-Service
  - A reward router dynamically selects reward function based on the query's task
  - Reward models are deployed as standalone services, accessible via HTTP

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Reasoning Benchmark	Metrics	MiMo-VL 7B-SFT	MiMo-VL 7B-RL	QVQ-72B Preview	Qwen2.5-VL 72B	Intern-VL3 78B	Qwen2.5 72B	GPT-4o	Gemini-2.5 Pro
Multi-modal									
OlympiadBench	Acc.	**59.4**	**59.4**	20.4	37.2*	12.3	-	25.9	69.8
MathVision	Acc.	57.9	**60.4**	35.9	38.1	43.2	-	31.2	69.1
$\text{MathVerse}^{\dagger}_{\text{vision_only}}$	Acc.	67.1	**71.5**	45.1*	57.6	51.0	-	49.9	76.7
DynaMath	Worst-case Acc.	**46.9**	45.9	30.7	38.1*	35.1	-	48.5	56.3
WeMath	Strict Score	65.1	**66.3**	37.7*	50.6*	46.1	-	50.6	78.0
LogicVista	Acc.	61.2	**61.4**	53.8*	57.1*	55.9	-	64.4	73.8
$\text{MathVista}^{\dagger}_{\text{mini}}$	Acc.	**81.8**	81.5	71.4	74.8	72.2	-	63.8	80.9
Text									
MATH500	Pass@1	95.0	**95.4**	83.8*	83.0	68.8*	82.8*	78.2*	95.2
AIME24	Pass@1	66.4	**67.5**	25.2*	16.7*	12.2*	19.4*	10.9*	92.0
AIME25	Pass@1	50.9	**52.5**	18.1*	10.8*	11.7*	13.3*	8.7*	86.7

Reasoning performance continued improved after RL

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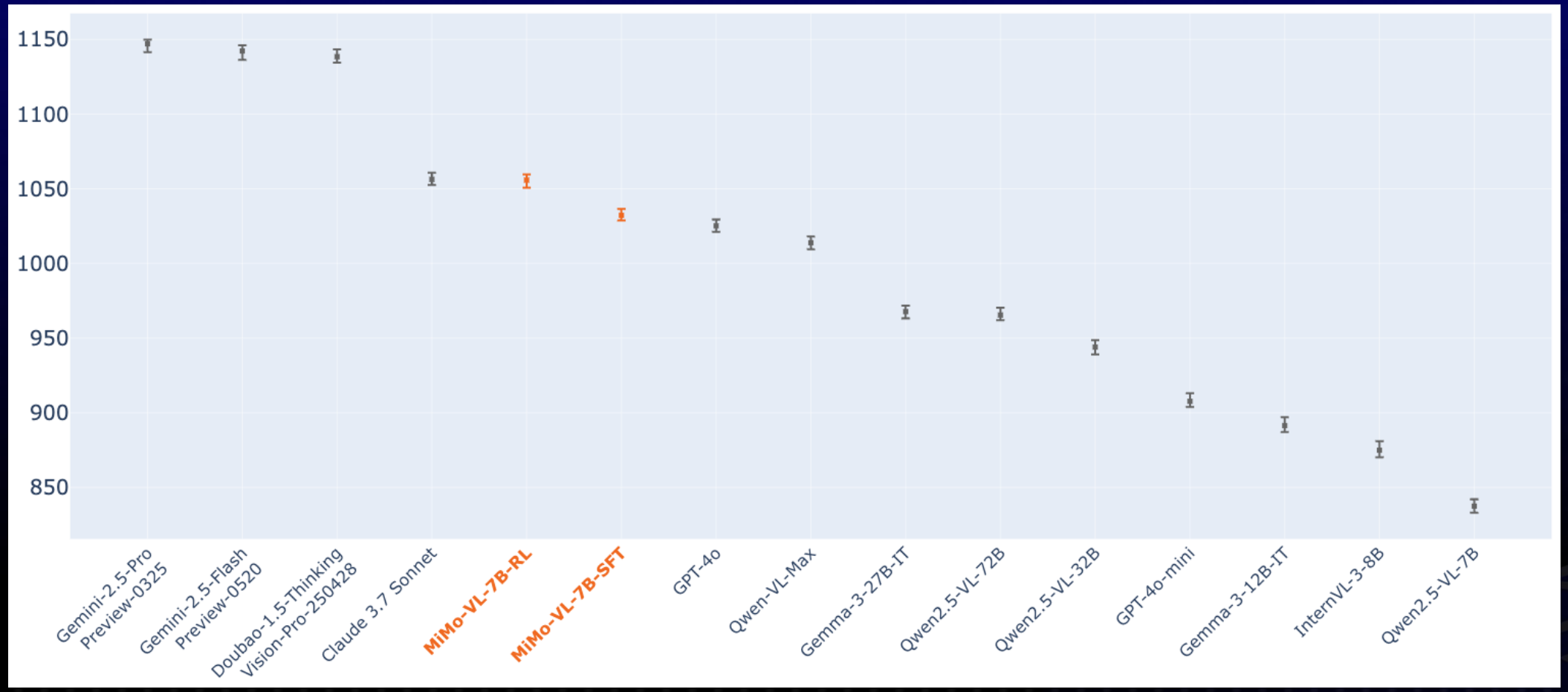

ELO rating continued improved after RL

永远相信，美好的事情即将发生。

加入「小米大模型Core」团队，
一起创造无限可能！

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- X (Twitter): https://x.com/XiaomiMiMo